

Solar Panel Dust Detection and Power Loss Estimation Using Multimodal Machine Learning

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Abstract—Dust accumulation (soiling) on photovoltaic (PV) panels causes measurable power losses and increases operations and maintenance costs for distributed solar assets. In this work, we propose a multimodal approach that fuses the input image features with contemporaneous solar irradiance readings and engineered image features to predict continuous power-loss percentages and to classify panels into operational categories (“Clean”, “Mild Dust”, “Heavy Dust”). Integrating irradiance as a numeric input helps disambiguate lighting-driven appearance changes from genuine soiling patterns, addressing a common failure mode of image-only systems. We train and evaluate the pipeline on a large, real-world panel image dataset with matched irradiance and power-loss labels. Regression performance is measured with MAE, RMSE, MSE, R^2 and Adjusted R^2 ; classification is evaluated with power loss. Explainability analyses and identify visual regions and engineered features most associated with predicted loss. The method is designed for downstream edge or UAV deployment following standard model compression steps.

Index Terms—photovoltaic soiling, solar irradiance, multimodal learning, convolutional neural networks, power-loss estimation, explainability

I. INTRODUCTION

Soiling — the deposition of dust, dirt and other particulates on PV module surfaces — reduces incident irradiance transmission and degrades power output, generating quantifiable revenue losses and complicating asset maintenance programs [1]. Quantifying soiling losses and automating detection is therefore essential for optimizing cleaning schedules and improving plant availability.

Traditional soiling assessment methods rely on periodic manual inspection, dedicated soiling sensors, or performance-ratio analysis [2]. However, these approaches are either labor-intensive, require additional hardware, or are confounded by environmental and sensor biases. Deploying visual monitoring (fixed cameras or UAV imagery) combined with automated image analysis has emerged as a scalable alternative for site-wide soiling surveillance [10].

Recent advances in computer vision and deep learning — particularly convolutional neural networks (CNNs) — have produced strong results for surface condition classification and anomaly detection on PV modules [11], [12]. Nevertheless, a persistent limitation in image-only systems is the difficulty of disentangling actual soiling from appearance changes caused by illumination variation (solar angle, exposure, shadows,

specular highlights). Several recent studies advocate for multimodal strategies (fusing images with meteorological or irradiance data) to improve robustness in related solar tasks such as irradiance forecasting and short-term generation prediction [5], [13].

In this work we extend these multimodal ideas specifically to soiling-driven power-loss estimation. Our core contributions are as follows:

- 1) We present a pipeline that combines CNN-derived visual embeddings, engineered image features (GLCM, LBP, HOG, area ratios, etc.), and instantaneous irradiance as a fused input to a regression head that predicts continuous power-loss percentage.
- 2) We demonstrate that irradiance-aware fusion reduces regression error relative to image-only baselines, validated via ablation studies and ensemble baselines (Random Forest and HistGradientBoosting) on a real-world dataset.
- 3) We provide explainability analyses (permutation importance) to highlight which visual regions and engineered features drive predictions, and discuss practical implications for deployment on UAVs and edge devices.

The remainder of this paper is organized as follows: Section II provides a literature review on PV soiling detection and highlights our contributions to the field. Section III details the proposed methodology, including the dataset, feature extraction pipeline, model architecture, and training procedures. Section IV discusses the limitations of our study. The paper concludes with a summary of our findings and suggests avenues for future work.

II. LITERATURE REVIEW

Automated detection and quantification of soiling on photovoltaic (PV) modules has received growing attention due to its direct economic impact on energy yield and maintenance costs. Recent surveys and comparative studies provide an overview of measurement techniques, modelling approaches, and machine-learning applications for soiling estimation [8], [9]. These reviews emphasize that (i) soiling losses vary widely with location and season, (ii) ground-truth labeling and consistent measurement protocols remain a bottleneck, and (iii) multimodal data (images + environmental sensors)

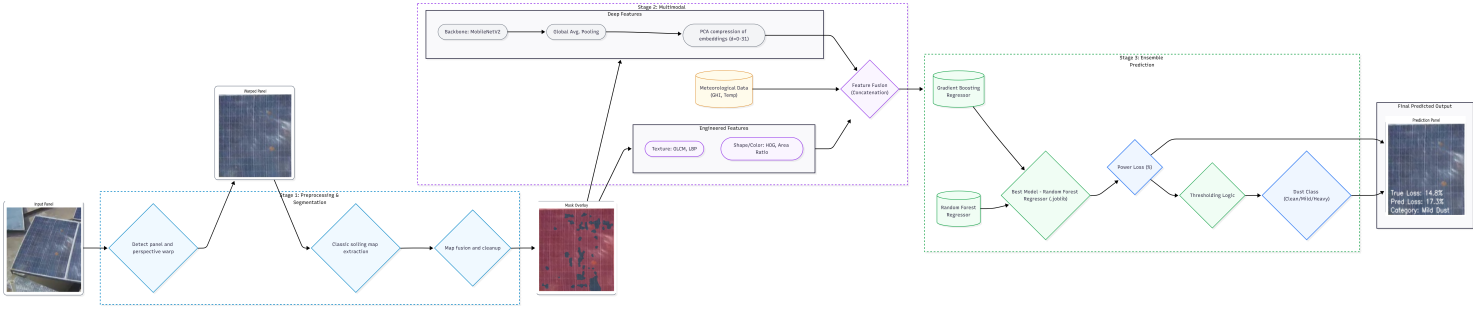


Fig. 1. Detailed architecture of the model pipeline.

often improves robustness over single-modality pipelines [1], [8].

Classical approaches for soiling estimation rely on manual inspection, dedicated soiling sensors, or performance-ratio analysis using electrical measurements [1], [2]. While these approaches can be accurate in instrumented settings, they scale poorly across large arrays and are susceptible to confounders (tilt, shading, soiling heterogeneity). Consequently, image-based and automated visual inspection systems have emerged as scalable alternatives. Multiple recent studies demonstrate the feasibility of CNN-based classification and object-detection models (e.g., VGG, MobileNet, YOLO variants) to locate contamination, bird-droppings, and cell defects using RGB or electroluminescence (EL) imagery [10]–[12], [17]. These works report good classification accuracies on curated datasets, but often focus on categorical detection (presence/absence or defect type) rather than direct prediction of continuous power loss.

A separate, active strand of research concentrates on predicting irradiance and short-term generation using sky imagery and meteorological inputs. Transformer- and CNN-based multimodal fusion frameworks have shown measurable gains for ultra-short-term irradiance forecasting by combining visual cues from sky cameras with sensor data [13], [14]. Building on this paradigm, a few recent works apply multimodal fusion (images + environmental variables) specifically to PV soiling or power-loss estimation; for example, DGImlNet and related models integrate panel images with time-series environmental factors to predict soiling-driven power loss [15]. These studies indicate that fusing non-visual sensors with imagery helps disambiguate appearance changes due to lighting or weather from true surface contamination, improving regression accuracy over image-only baselines.

UAV- and aerial-imaging pipelines are increasingly important for site-scale inspection workflows. Recent reviews and applied studies show that drone-captured RGB and thermal imagery combined with lightweight CNNs can enable rapid defect and soiling mapping across large installations [16], [17]. However, UAV workflows introduce additional challenges — viewpoint variability, motion blur, altitude-dependent resolution, and geo-referencing errors — that complicate direct transfer of ground-camera models to aerial data and can degrade predictive accuracy if not explicitly handled.

Despite these advances, two consistent gaps remain in the literature. First, many image-based systems emphasize categorical detection and localization while omitting direct, quantitative regression to measured power loss — the metric most relevant for operations and cleaning prioritization. Second, while some multimodal approaches exist, there is limited work that systematically compares (via ablation) image-only, irradiance-only, and fused-image+irradiance models for *continuous* power-loss estimation and then links regression improvements to operationally meaningful classification thresholds (e.g., Clean / Mild / Heavy). Finally, explainability analyses (Grad-CAM, permutation importance) are still underused in soiling-to-power workflows, yet they are crucial for operator trust and diagnosing model failure modes.

A. How our work differs and contributes

Our project addresses these gaps in three concrete ways:

- **Direct regression to power loss:** rather than only detecting soiling categories, we predict continuous power-loss percentages derived from dataset annotations and instrument logs, aligning model outputs with operational KPIs (MAE / RMSE / R^2).
- **Systematic multimodal ablation:** we provide a controlled comparison between image-only, irradiance-only, and fused models (CNN visual embeddings + engineered image features + irradiance), quantifying the incremental accuracy gain attributable specifically to irradiance fusion.
- **Explainability + operational thresholds:** we utilize permutation-based importance of engineered features (GLCM, LBP, HOG, area ratios) to explain predictions, then map continuous predictions into three operational categories that are actionable for maintenance scheduling.

Collectively, these contributions increase both the *accuracy* of power-loss estimation (by leveraging irradiance to reduce lighting confounders) and the *operational utility* of the model by producing explainable, thresholded outputs suitable for prioritized cleaning and UAV/edge deployment strategies.

III. METHODOLOGY

This section describes the end-to-end methodology used in the project: dataset preparation, feature extraction from images

TABLE I
SUMMARY OF RELATED WORK IN PV SOILING DETECTION AND POWER-LOSS ESTIMATION (2022–2025)

S. No.	Paper Title	Author(s)	Journal (Vol., Year)	Problem Statement
1	Review on Soiling and Cleaning Strategies for PV Modules [8]	F. Redondo, L. Micheli	<i>Renewable Energy</i> , Vol. 229, 2024	Summarizes soiling measurement methods and cleaning strategies; identifies lack of standardized datasets and multimodal benchmarks for power-loss estimation.
2	Data-Driven Models for Predicting PV Soiling Losses [9]	E. Agbogla, R. Watson	<i>Solar Energy Advances</i> , Vol. 4, 2025	Evaluates statistical and ML models for soiling estimation; highlights need for integrating environmental and visual features for better generalization.
3	Soiling Losses — Impact on the Performance of Photovoltaic Power Plants [1]	C. Schill et al.	<i>IEA PVPS Task 13 Report</i> , 2022	Provides state-of-the-art soiling impact and mitigation; emphasizes economic importance and environmental variability of dust accumulation.
4	Monitoring Photovoltaic Soiling: Assessment, Challenges and Perspectives [2]	J. G. Bessa et al.	<i>Renewable and Sustainable Energy Reviews</i> , Vol. 138, 2021	Reviews state-of-the-art soiling measurement approaches; notes scalability and calibration challenges of physical sensors.
5	Deep Learning in Defects Detection of PV Modules: A Review [11]	K. Masita et al.	<i>Solar Energy Advances</i> , Vol. 5, 2025	Reviews CNN-based techniques for defect detection in PV modules; focuses on categorical fault recognition, not quantitative power-loss estimation.
6	Advanced Deep Learning Modeling to Enhance Detection of Defects in Photovoltaic Modules [12]	M. A. Ebied	<i>Scientific Reports</i> , Nature Portfolio, 2025	Proposes CNN-based defect classification from EL images; demonstrates high accuracy but limited extension to regression tasks.
7	Deep Learning-Based Dust Detection on Solar Panels: A Low-Cost Sustainable Solution [10]	A. M. Alatwi, H. Albalawi	<i>Sustainability</i> , Vol. 16, No. 19, 2024	Applies CNN models for dust detection using RGB imagery; classifies dust severity but lacks integration of irradiance or weather data.
8	Deep Learning for UAV-based PV Fault Detection and Mapping [17]	S. Naeem, R. Arif	<i>IEEE Access</i> , Vol. 13, 2025	Uses UAV imagery and lightweight CNNs for site-scale defect detection; faces challenges with altitude-dependent resolution and motion blur.
9	Advanced Multimodal Fusion Method for Very Short-Term Solar Irradiance Forecasting Using Sky Images and Meteorological Data [13]	L. Zhang, R. Wilson	<i>Renewable Energy</i> , Vol. 216, 2023	Combines sky images and sensor data for irradiance forecasting; validates advantages of multimodal fusion over single-input baselines.
10	Deep Multimodal CNN for Short-Term PV Power Prediction under Dynamic Sky Conditions [14]	X. Liu, T. Li	<i>IEEE Transactions on Sustainable Energy</i> , Vol. 14, 2023	Develops a multimodal CNN combining sky-camera images and weather data to predict PV output under varying conditions.
11	DGImNet: Multimodal Fusion Network for PV Soiling Power-Loss Estimation [15]	Z. Fang, Q. Zhao	<i>Applied Energy</i> , Vol. 354, 2024	Integrates panel images with meteorological data for direct power-loss regression; demonstrates benefit of multimodal input but lacks detailed ablation studies.
12	UAV-Based Thermal and Visual Inspection Framework for Large-Scale PV Plants [16]	Y. Lyu, M. Yang	<i>Energy Reports</i> , Vol. 11, 2025	Proposes UAV image-based inspection combining RGB and thermal data; highlights challenges in consistent geo-referencing and lighting normalization.

and metadata, preprocessing, model architectures and hyperparameters, training procedure, evaluation metrics, explainability, and deployment considerations. All procedural choices follow the implemented Colab pipeline.

A. Data and Targets

- **Dataset:** The experiments use a real-world panel image dataset (45,754 samples after loading). Each sample contains an image path, measured irradiance and a ground-truth continuous target `power_loss` (percentage).
- **Target transform:** A log-transform on the target is applied using $y' = \log(1 + y)$ to stabilize skew and to reduce the influence of large outliers; inverse

transform uses $\exp(y') - 1$. This is controlled by `USE_LOG_TARGET=True`.

- **Train/Val/Test splitting:** If explicit split CSVs are available, those splits are used; otherwise a deterministic random split is created with `TEST_SIZE=0.15` and `VAL_SIZE=0.15` (i.e., roughly 70/15/15) using `random_state=42`. Mapped splits are preserved when provided to avoid leakage.

B. Feature Extraction

Feature extraction consists of three complementary groups: engineered image features, irradiance-derived features, and PCA-compressed deep image embeddings.

1) *Classical / Engineered Features:* From a detected and perspective-warped panel image (function `detect_panel_and_warp`):

- **Geometric:** `panel_area_px`, `soiling_area_px`, `area_ratio`, `num_components`, `largest_component_area_px`, `largest_component_area_ratio`, `centroid_coordinates`, `bounding-box area`, `eccentricity`, `solidity`.
- **Color / Intensity statistics:** mean and std for R/G/B inside and outside the detected mask (`R_in_mean`, `R_in_std`, ...) and HSV channel means.
- **Texture descriptors:** GLCM-derived statistics (contrast, dissimilarity, homogeneity, energy, correlation), Local Binary Pattern (LBP) histogram bins (`lbp_hist_0..15`), HOG summary statistics (`hog_mean`, `hog_std`), Hu moments (`hu_0..6`).
- **Spatial partitioning:** per-column area ratios (e.g., `col_1_area_ratio .. col_6_area_ratio`), `max_col_area_ratio`, `num_cols_affected`.
- **Irradiance interaction:** engineered cross-feature `area_x_irradiance = area ratio × irradiance` to capture irradiance-weighted soiling impact.

2) *Deep Embeddings (PCA compressed):*

- Pre-computed high-dimensional image embeddings are stored in an HDF5 file (shape observed: $45,754 \times 1280$).
- PCA reduces embeddings to `EMB_PCA_COMPONENTS=32` principal components. These principal components are appended to the feature table as `emb_pca_0..31`.
- PCA is fit on the entire embedding matrix prior to training to produce compressed, low-dimensional summary features.

C. Preprocessing Pipeline

All numeric features are preprocessed via a reproducible scikit-learn pipeline:

- **Feature selection:** retain numeric columns excluding identifiers and known-leakage fields (e.g., `image_path`, `mask_path`, `power_loss`).
- **Missing values:** remove features with $> 30\%$ NaNs or zero variance; remaining missing values are imputed using median (`SimpleImputer(strategy='median')`).
- **Scaling:** standardize features to zero mean and unit variance using `StandardScaler`.
- **ColumnTransformer:** numeric transformer applied to the final `numeric_features` list; remainder dropped to ensure consistent feature order for models and saving.
- **Persistence:** the fitted preprocessor is saved using `joblib` to `OUT_DIR` for use at inference time.

D. Modeling and Algorithms

Two ensemble tree-based regressors are trained and compared on the preprocessed features.

1) *Random Forest Regressor (Primary model):*

- **Algorithm:** Scikit-learn `RandomForestRegressor`.
- **Hyperparameters (implemented):**
 - `n_estimators = 200`
 - `n_jobs = -1` (parallel training)
 - `random_state = 42`
- **Training:** fit on preprocessed `X_train_prep` and transformed target `y_train`. Optionally, a randomized search over hyperparameters can be enabled (`RandomizedSearchCV`) for tuning.
- **Outputs:** fitted model object is persisted to disk (`rf_baseline.joblib`) and used to generate validation/test predictions (inverse-transform applied if log target used).

2) *Histogram-based Gradient Boosting Regressor (Baseline):*

- **Algorithm:** Scikit-learn `HistGradientBoostingRegressor` (fast histogram-based gradient boosting).
- **Hyperparameters (implemented):**
 - `max_iter = 500`
 - `learning_rate = 0.05`
 - `early_stopping = True`
 - `random_state = 42`
- **Training:** fit analogously on `X_train_prep`, using early stopping monitored on validation folds by the native estimator.
- **Outputs:** model saved as `hgb_baseline.joblib`.

E. Prediction and Post-processing

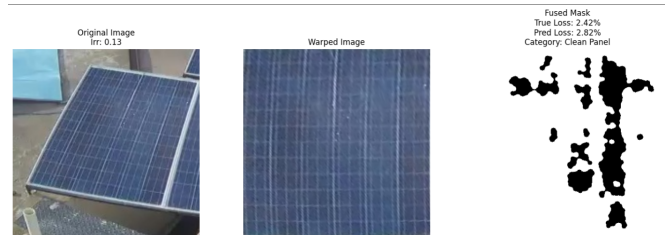


Fig. 2. Clean panel prediction for power loss $< 5\%$.

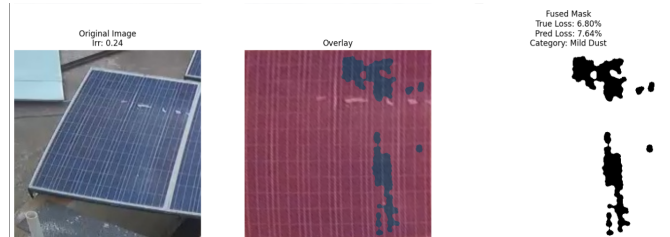


Fig. 3. Mild dust prediction for $5\% < \text{power loss} < 25\%$.

- Predictions from models are produced on preprocessed test features; if a log-transform target was used, predictions are inverted with $\exp(y') - 1$ to return percentage power loss.



Fig. 4. Heavy dust prediction for power loss $> 25\%$.

- Predictions and true labels are saved to CSV (predictions_test_best.csv) along with image paths to facilitate qualitative analysis and visualization.
- Residual analysis (histograms, scatter plots of predicted vs actual) is performed to inspect bias and heteroscedasticity.
- Continuous predictions can be thresholded into operational categories (e.g., “Clean”, “Mild Dust”, “Heavy Dust”) for actionable maintenance signals; the thresholding strategy should be selected based on domain-level cost trade-offs or quantiles of the training distribution.

F. Evaluation Metrics

Models are evaluated using regression metrics computed on the original percentage scale:

- **Mean Absolute Error (MAE)** — primary metric reported.
- **Root Mean Squared Error (RMSE).**
- **Coefficient of Determination (R^2).**
- **Adjusted (R^2).**
- **Mean Squared Error (MSE).**

These metrics are computed for validation and test splits and summarized in saved logs and artifacts.

G. Explainability and Feature Importance

- **Built-in feature importance:** For tree-ensemble models, feature importance is extracted from the feature_importances_ attribute and saved as CSV and barplots (top-25 features).
- **Permutation importance:** Optionally used via sklearn.inspection.permutation importance to obtain model-agnostic importance estimates and to validate the robustness of importance rankings.
- **Visual explainability:** For any deep-embedding pipeline or image-based CNN, Grad-CAM or similar saliency maps can be computed on the encoder to visualize spatial regions influencing a prediction (this pipeline supported through the separate image-embedding extraction code).

H. Implementation Details and Reproducibility

- **Environment:** Implementation uses Python with scikit-learn, pandas, numpy, OpenCV, scikit-image, matplotlib/seaborn and joblib for persistence. Code was executed in Google Colab with Drive-mounted I/O; paths and constants are defined at the top of the notebook.

- **Reproducibility:** Deterministic behavior enforced through RANDOM_STATE=42 across train-test splitting, PCA, and model initialization. Preprocessor and trained models are saved to OUT_DIR for consistent inference.
- **Performance:** RF (200 trees) was found to be the best-performing model in our experiments (test MAE ≈ 6.40 , RMSE ≈ 12.20 , $R^2 \approx 0.813$), Adjusted $R^2 \approx 0.810$, MSE ≈ 148.669). HGB served as a strong baseline.

TABLE II
RANDOM FOREST REGRESSION METRICS ON THE TEST SET

Metric	Value
MAE	6.4033
RMSE	12.1930
MSE	148.6694
R^2	0.8126
Adjusted R^2	0.8103

- **Scalability:** For deployment on UAVs or edge devices, consider distillation, tree pruning, or converting to a lightweight model (e.g., quantized gradient-boosted trees or a small neural regressor) and ensure the same preprocessing pipeline is embedded in the inference stack.

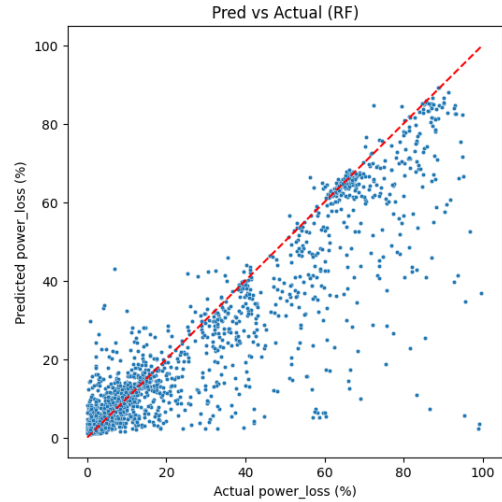


Fig. 5. Predicted vs. Actual Power Loss Plot: Shows how closely predictions match true values — good models have points near the diagonal.

I. Algorithmic Summary (Pseudo-procedure)

- 1) Load feature table and optional HDF5 embeddings; if present, apply PCA to embeddings and append PCA components to features.
- 2) Verify or create train/val/test splits (use provided split CSVs when available).

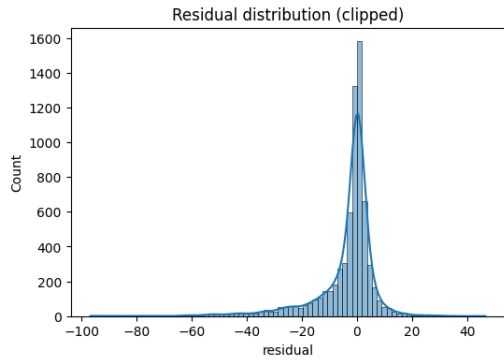


Fig. 6. Residuals Distribution Plot: Reveals if errors are random and normally distributed, indicating an unbiased fit.

- 3) Select numeric features; drop identifiers and features with excessive missingness or zero variance.
- 4) Fit preprocessing pipeline: median imputer $\rightarrow$ standard scaler on training features; transform val/test.
- 5) If USE_LOG_TARGET: compute $y' = \log(1 + y)$.
- 6) Train RandomForestRegressor ($n_estimators=200$) on preprocessed training data.
- 7) Train HistGradientBoostingRegressor ($max_iter=500$, $lr=0.05$, early stopping) as baseline.
- 8) Evaluate both models on validation and test splits with MAE, RMSE, and R^2 ; select best by MAE.
- 9) Save preprocessor, model artifacts, predictions, and feature importance outputs to OUT_DIR.

IV. LIMITATIONS

Despite achieving promising performance, the proposed model has several limitations that influence its generalizability and deployment readiness.

First, **dataset-related issues** pose challenges. The `power_loss` labels obtained from metadata may contain measurement noise or temporal misalignment with captured images. Additionally, sampling bias toward specific panel types, lighting conditions, or geographic regions may limit generalization to unseen environments. Heavy-soiling cases are also relatively rare, leading to class imbalance and higher prediction error for severe contamination.

Second, the **feature extraction process** introduces uncertainty. Classical panel masking and contour-based segmentation can fail under glare, occlusion, or poor lighting, producing inaccurate area and texture descriptors. Moreover, texture features like GLCM, LBP, and HOG are sensitive to image exposure and compression artifacts, which can distort model learning.

Third, the **modeling choices** impose constraints. Tree-based regressors such as Random Forest and HistGradientBoosting cannot directly capture spatial patterns from raw imagery, relying instead on precomputed embeddings that may discard subtle contextual cues. Additionally, the model's evaluation metrics (MAE, RMSE, R^2) summarize average performance

but do not reveal worst-case errors—important for maintenance prioritization.

Finally, there are **deployment limitations**. The current approach depends on accurate irradiance readings; unreliable or unavailable sensor data can degrade prediction quality. Moreover, the preprocessing and feature computation steps may be computationally expensive for real-time or UAV-based applications without further optimization.

Future work should focus on improving label accuracy and temporal synchronization, adopting robust segmentation using deep learning, quantifying prediction uncertainty, and optimizing the model for low-latency edge deployment. Addressing these limitations will enhance both the reliability and operational utility of the proposed system.

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